

Exercises from Ch. 5: Submanifolds, Lee

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Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

Exercises

Exercise 0.1. (Exercise 1)

Consider the map $\Phi: \mathbb{R}^4 \rightarrow \mathbb{R}^2$ defined by

$$\Phi(x, y, s, t) = (x^2 + y, x^2 + y^2 + s^2 + t^2 + y).$$

Show that $(0, 1)$ is a regular value of Φ , and that the level set $\Phi^{-1}(0, 1)$ is diffeomorphic to $\mathbb{S}^2$.

Proof. The level set $\Phi^{-1}(0, 1) \subseteq \mathbb{R}^4$ consists of $(x, y, s, t) \in \mathbb{R}^4$ such that $x^2 + y = 0$ and $x^2 + y^2 + s^2 + t^2 + y = 1$. Hence, $y = -x^2$, so $y \leq 0$ and

$$\Phi^{-1}(0, 1) = \{(x, -x^2, s, t) \in \mathbb{R}^4 : x^4 + s^2 + t^2 = 1\}.$$

We show that for each $p \in \Phi^{-1}(0, 1)$ the differential map

$$d\Phi_p: T_p\mathbb{R}^4 \rightarrow T_{\Phi(p)}\mathbb{R}^2 = T_{(0,1)}\mathbb{R}^2$$

is surjective (note that we have the following identifications: $T_p\mathbb{R}^4 \cong \mathbb{R}^4$ and $T_{(0,1)}\mathbb{R}^2 \cong \mathbb{R}^2$). Let $p = (x, y, s, t) \in \Phi^{-1}(0, 1)$, i.e. $y = -x^2$ (or equivalently $x = \pm\sqrt{-y}$), $y \leq 0$, and $(y, s, t) \in \mathbb{S}^2$. Let Φ^1 and Φ^2 denote the component functions of $\Phi = (\Phi^1, \Phi^2)$. The Jacobian of Φ at p is given by

$$D\Phi_p = \begin{pmatrix} \frac{\partial \Phi^1}{\partial x} & \frac{\partial \Phi^1}{\partial y} & \frac{\partial \Phi^1}{\partial s} & \frac{\partial \Phi^1}{\partial t} \\ \frac{\partial \Phi^2}{\partial x} & \frac{\partial \Phi^2}{\partial y} & \frac{\partial \Phi^2}{\partial s} & \frac{\partial \Phi^2}{\partial t} \end{pmatrix} = \begin{pmatrix} 2x & 1 & 0 & 0 \\ 2x & 2y & 2s & 2t \end{pmatrix}.$$

We want to show that $\text{rank}(D\Phi_p) = \dim T_{(0,1)}\mathbb{R}^2 = 2$. If $x \neq 0$ ($y \neq 0$), then the submatrix

$$\begin{pmatrix} 2x & 1 \\ 2x & 2y \end{pmatrix}$$

has a nonzero determinant. If $x = 0$ ($y = 0$), then $p \in \Phi^{-1}(0, 1)$ implies that $s^2 + t^2 = 1$. Then if $s \neq 0$, then submatrix

$$\begin{pmatrix} 1 & 0 \\ 2y & 2s \end{pmatrix}$$

has a nonzero determinant and if $s = 0$, then $t = \pm 1$ and the submatrix

$$\begin{pmatrix} 1 & 0 \\ 2y & 2t \end{pmatrix}$$

has a nonzero determinant. In any case, i.e. for every p ,

$$\text{rank}(D\Phi_p) = \text{rank}(d\Phi_p) = \dim T_{(0,1)}\mathbb{R}^2 = 2.$$

Therefore $d\Phi_p$ is surjective for all $p \in \Phi^{-1}(0, 1)$. Hence, each $p \in \Phi^{-1}(0, 1)$ is a regular point of Φ and thus $(0, 1)$ is a regular value of Φ as desired. Additionally, $\varphi^{-1}(0, 1)$ is a regular level set and $\Phi: \mathbb{R}^4 \rightarrow \mathbb{R}^2$ is a smooth map between smooth manifolds, so by the Regular Level Set Theorem (Corollary, 5.14), $\Phi^{-1}(0, 1) \subseteq \mathbb{R}^4$ is a properly embedded submanifold of codimension equal to $\dim \mathbb{R}^2 = 2$. Thus, $\dim \Phi^{-1}(0, 1) = 2$.

We now prove that $\Phi^{-1}(0, 1) \cong \mathbb{S}^2$

Consider the equivalent condition for a point $p \in \mathbb{R}^4$ to be in the level set $\Phi^{-1}(0, 1)$, given by substituting $y = -x^2$ ($x^2 + y = 0$) into $x^2 + y^2 + s^2 + t^2 + y = 1$; we have the following identification. Let $\iota: \Phi^{-1}(0, 1) \rightarrow \iota(\Phi^{-1}(0, 1)) = M$ be given by $(x, y, s, t) \mapsto (x, s, t)$ where $x^4 + s^2 + t^2 = 1$. Hence, $M = \{(x, s, t) \in \mathbb{R}^3 : x^4 + s^2 + t^2 = 1\}$. Clearly, ι is smooth. It is bijective since $y = -x^2$ and $\iota^{-1}: M \rightarrow \Phi^{-1}(0, 1)$ given by $(x, s, t) \mapsto (x, -x^2, s, t)$ is also smooth. Hence ι is a diffeomorphism, so $\Phi^{-1}(0, 1) \cong M$. Therefore, it suffices to show that $M \cong \mathbb{S}^2$.

Define

$$\psi: \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad (x, s, t) \mapsto \left(x, \frac{s}{\sqrt{1+x^2}}, \frac{t}{\sqrt{1+x^2}} \right).$$

This map is well-defined and smooth since the map $x \mapsto \frac{1}{\sqrt{1+x^2}}$ is smooth ($1+x^2 \geq 1$) and the component functions ψ_s, ψ_t are compositions of smooth maps. We show that $\psi|_M: M \rightarrow \mathbb{S}^2$ is a diffeomorphism ($\psi|_M$ is smooth since $M \subseteq \mathbb{R}^3$). For every $(x, s, t) \in M$, $\psi(x, s, t) \in \psi(M) \subseteq \mathbb{S}^2$. We first prove that $\psi|_M$ is bijective onto $\mathbb{S}^2$. Let $(a, b, c) \in \mathbb{S}^2$. Setting $x = a$, $s = b\sqrt{1+a^2}$, and $t = c\sqrt{1+a^2}$, we have $\psi(x, s, t) = (a, b, c)$ and $(x, s, t) \in M$ since $x^4 + s^2 + t^2 = a^4 + b^2(1+a^2) + c^2(1+a^2) = a^4 + (1+a^2)(b^2 + c^2) = a^4 + (1+a^2)(1-a^2) = 1$. Hence $\psi|_M$ is surjective and $\mathbb{S}^2 = \psi(M)$. Injectivity follows immediately since for any two points $(x, s, t), (x', s', t') \in M$, $\psi(x, s, t) = \psi(x', s', t')$ implies that $x = x'$ and therefore $s = s'$ and $t = t'$. The inverse map

$$\psi|_M^{-1}: \mathbb{S}^2 \rightarrow M$$

is given by $(a, b, c) \mapsto (a, b\sqrt{1+a^2}, c\sqrt{1+a^2})$ which is clearly smooth. Thus, $\psi|_M$ is a diffeomorphism. Therefore, $M \cong \mathbb{S}^2$ and in particular, $\Phi^{-1}(0, 1) \cong \mathbb{S}^2$. □